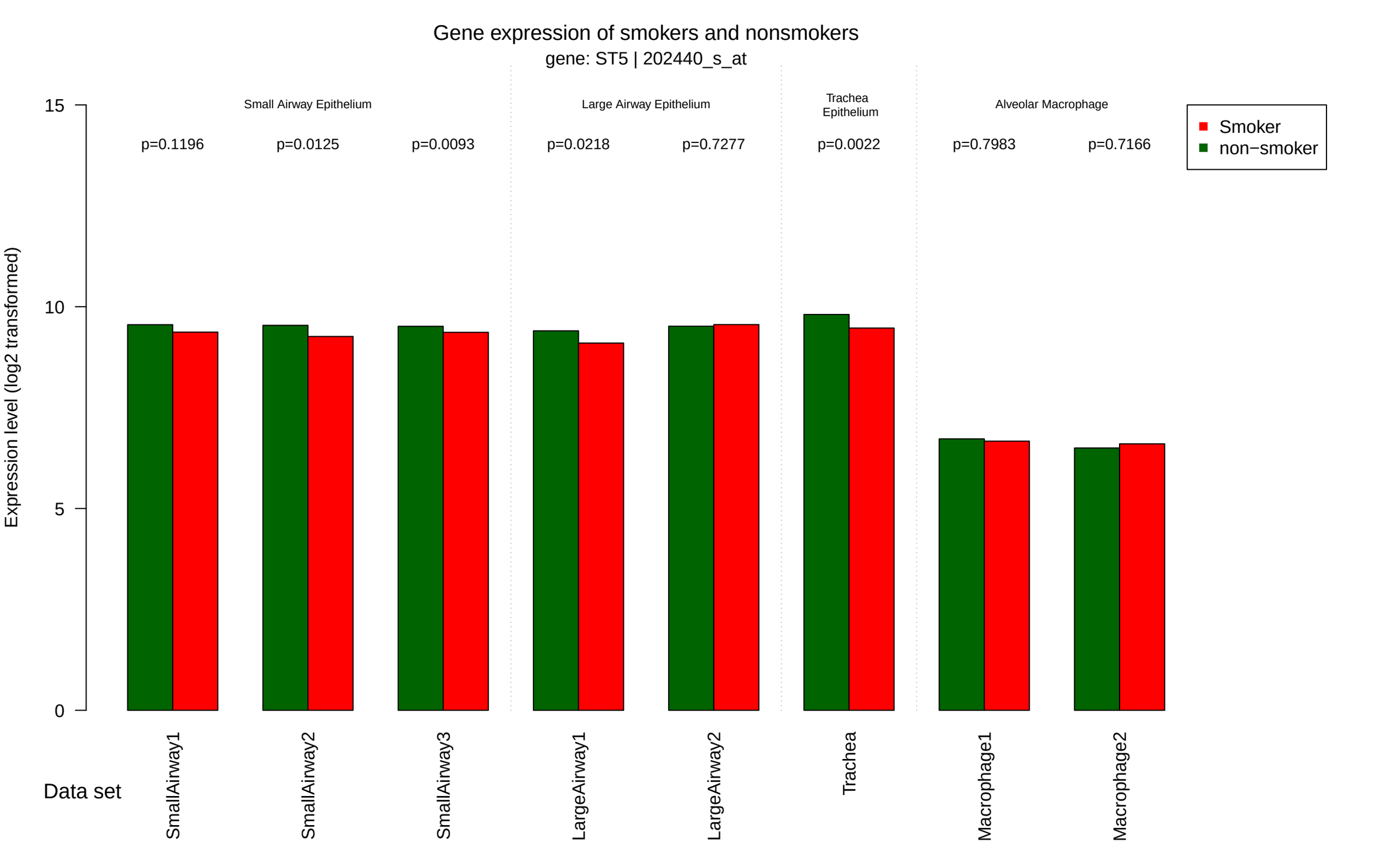
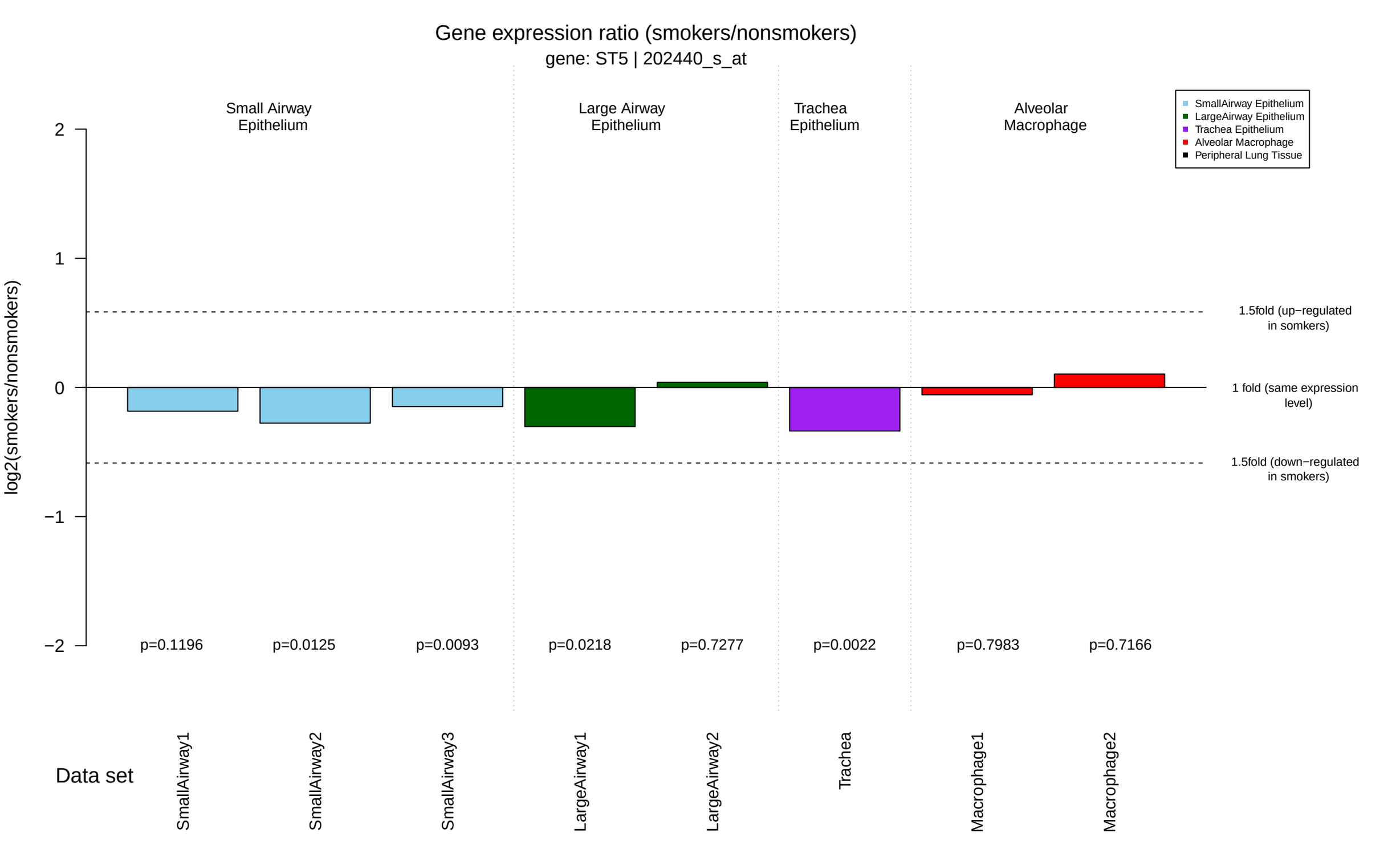
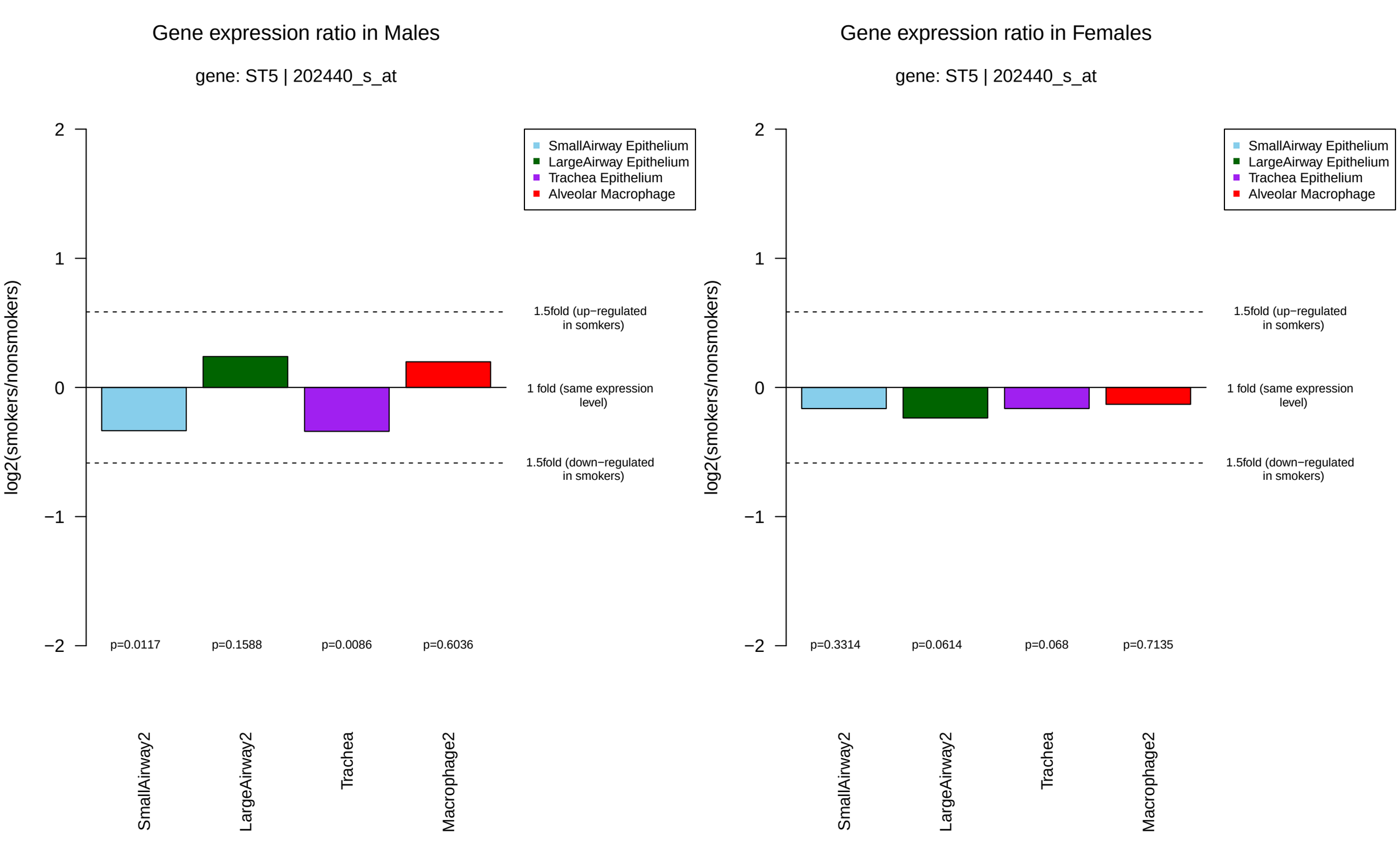
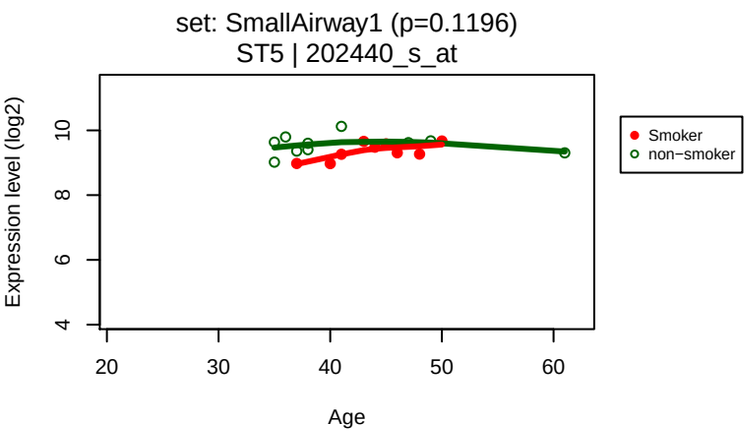
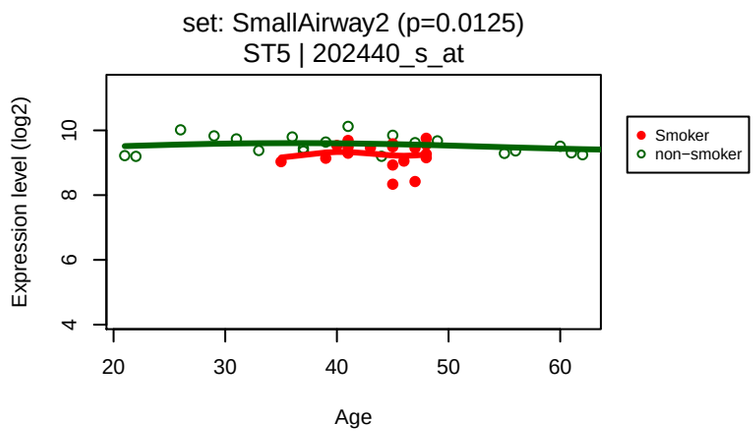
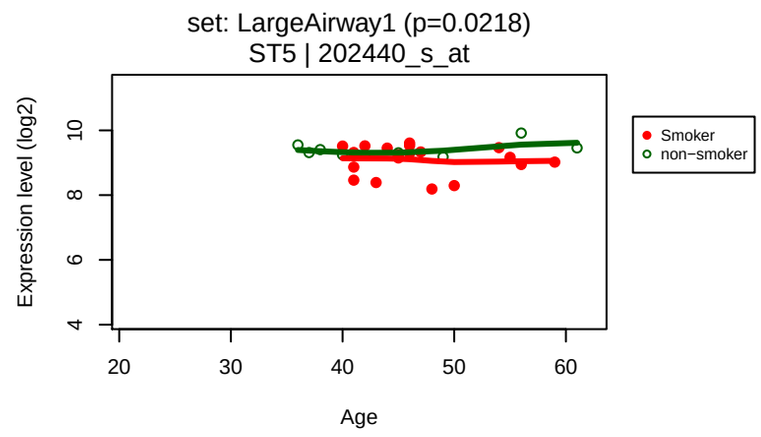

Gene expression of smokers and nonsmokers
gene: ST5 | 202440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1196
p=0.0125
p=0.0093
p=0.0218
p=0.7277
p=0.0022
p=0.7983
p=0.7166
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

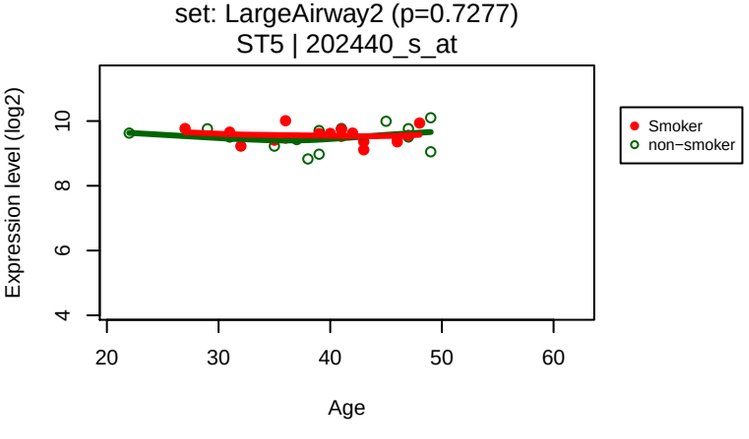

Gene expression ratio (smokers/nonsmokers)
gene: ST5 | 202440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1196
p=0.0125
p=0.0093
p=0.0218
p=0.7277
p=0.0022
p=0.7983
p=0.7166
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

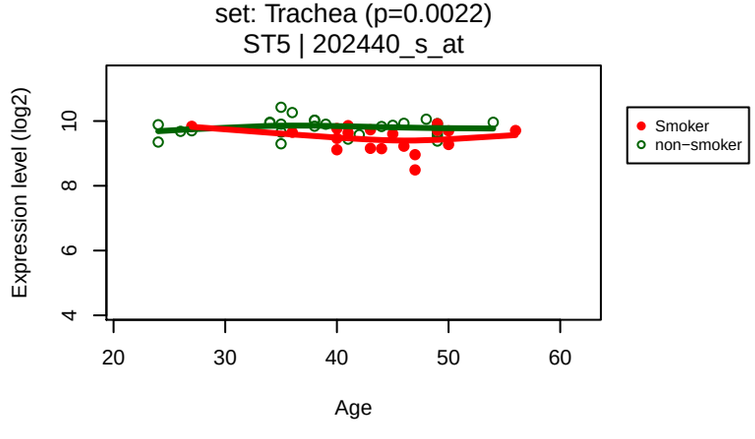

Gene expression ratio in Males
gene: ST5 | 202440_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0117
p=0.1588
p=0.0086
p=0.6036
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST5 | 202440_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3314
p=0.0614
p=0.068
p=0.7135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

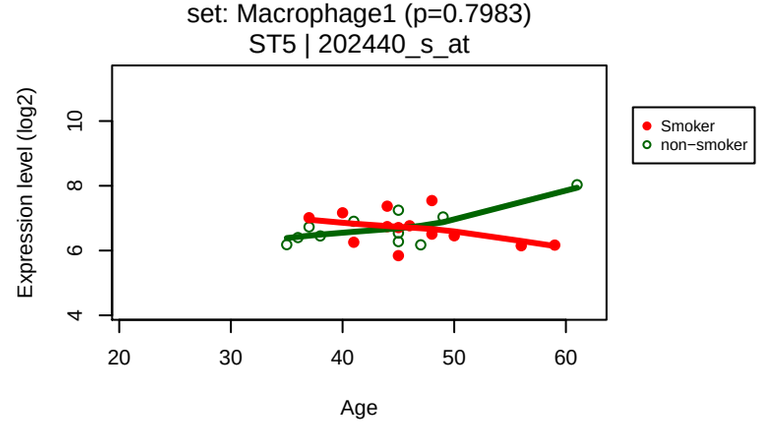

set: SmallAirway1 (p=0.1196)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker

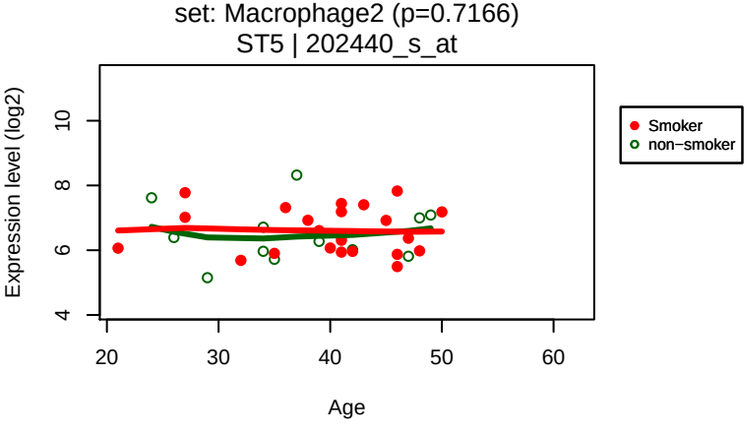

set: SmallAirway2 (p=0.0125)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker

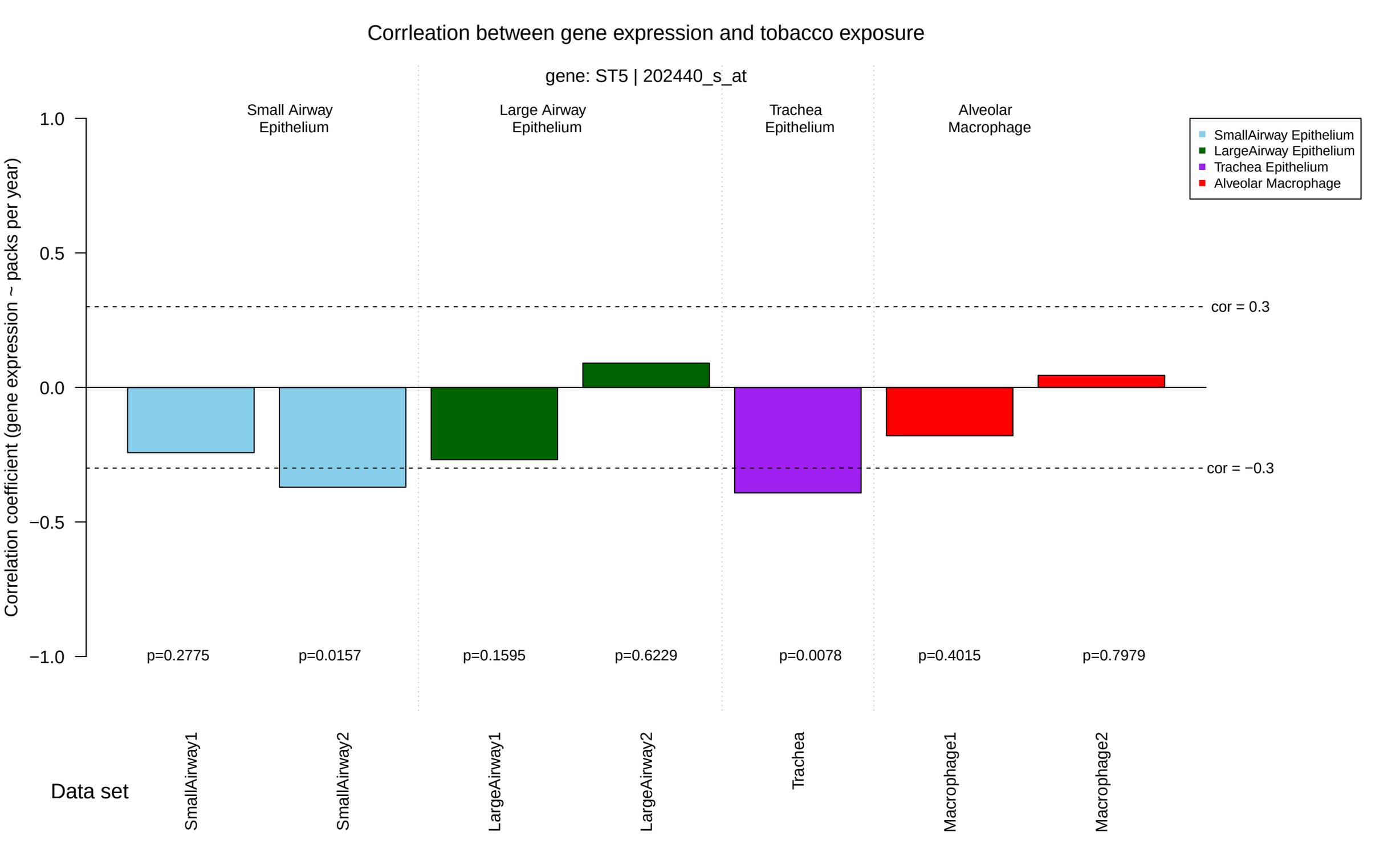

set: LargeAirway1 (p=0.0218)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7277)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7983)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7166)
ST5 | 202440_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST5 | 202440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2775
p=0.0157
p=0.1595
p=0.6229
p=0.0078
p=0.4015
p=0.7979
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2